Training Day – 11

Today's training focused on **Git and GitHub** and their role in version control, collaborative development, and project management. The session provided extensive hands-on experience with repositories, commits, branches, remote synchronization, and the fork workflow.

Work Undertaken:

• Learned the fundamentals of **Git** as a **Distributed Version Control System (DVCS)** and understood how it helps developers track changes and manage different versions of a project.

• Understood the difference between **Git and GitHub**. Git is a version control system used locally, while GitHub is a cloud-based platform used to host Git repositories and support collaboration.

• Created **local and remote repositories** and learned how to connect them for synchronizing project changes.

• Learned how to **create, switch, and merge branches**, allowing different features or changes to be developed independently without affecting the main project.

• Practiced the complete **fork and synchronization workflow** by forking a repository, connecting it to a local repository, making changes, and pushing updates.

• Completed hands-on Git projects by creating websites, managing different branches, merging changes, pulling the latest updates from GitHub, and viewing the **commit history**.

Additional Learning:

The practical exercises demonstrated how Git supports organized and collaborative software development. Working with branches, remote repositories, forks, and synchronization helped me understand how developers can work on different features simultaneously while maintaining a structured project history.

Conclusion:

The day provided extensive practical knowledge of **Git and GitHub**. By completing multiple hands-on exercises involving repositories, branches, commits, merges, and remote synchronization, I developed a stronger understanding of version control and collaborative development workflows used in real-world software projects.